

Scala For DS

Practical no:-25

Roll No. : S019	Name: SUMEET KUMAVAT
Class : SY BSC IT	Date/Time of Submission : 01-09-2026

Aim:- Handle missing values in a dataset. Replace missing values with the column mean.

CODE:-

```
build.sbt  Practical25.scala x
1  object Practical25 extends App { new *
2
3  val sales: List[java.lang.Double] = List(100.0, 200.0, null, 300.0, null)
4  val v = sales.filter(_ != null)
5  val mean = v.map(_.toDouble).sum / v.length
6  println("Mean Sales: " + mean)
7
8  val n = sales.map { value =>
9    if (value == null) mean
10   else value
11 }
12
13 println("Original Data:")
14 println(sales.mkString(", "))
15
16 println("After Replacing Missing Values:")
17 println(n.mkString(", "))
18 println("\nSumeet S019")
19 }
```

OUTPUT:-

```
"C:\Program Files\Java\jdk1.8.0_181\bin\java.exe" ...
```

```
Mean Sales: 200.0
```

```
Original Data:
```

```
100.0, 200.0, null, 300.0, null
```

```
After Replacing Missing Values:
```

```
100.0, 200.0, 200.0, 300.0, 200.0
```

```
Sumeet S019
```

```
Process finished with exit code 0
```

Scala For DS

Practical no:-26

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Aim:-

CODE:-

```
build.sbt  Practical25.scala x
1  import breeze.linalg._
2  ▶ object Practical25 extends App { new *
3
4      val x = DenseVector(1.0, 2.0, 3.0, 4.0, 5.0) // Dataset
5      val y = DenseVector(0.0, 0.0, 0.0, 1.0, 1.0) // Binary labels
6
7      // Initial weight and bias
8      var weight = 0.0
9      var bias = 0.0
10
11     // Sigmoid function
12     def sigmoid(z: Double): Double = { new *
13         1.0 / (1.0 + math.exp(-z))
14     }
15
16     // Train the model
17     for (i <- 1 to 1000) {
18         for (j <- 0 until x.length) {
19
20             val prediction = sigmoid(weight * x(j) + bias)
21
22             val error = prediction - y(j)
23
24             weight = weight - 0.01 * error * x(j)
25             bias = bias - 0.01 * error
26         }
27     }
```

```

25         bias = bias - 0.01 * error
26     }
27 }
28
29 // Classify the data
30 println("Predictions:")
31
32 for (i <- 0 until x.length) {
33
34     val probability = sigmoid(weight * x(i) + bias)
35     val result =
36         if (probability >= 0.5) 1 else 0
37
38     println("Input = " + x(i) +
39           ", Probability = " + probability +
40           ", Class = " + result)
41 }
42 println("\nSumeet S019")
43 }

```

OUTPUT:-

```

"C:\Program Files\Java\jdk1.8.0_181\bin\java.exe" ...
Predictions:
Input = 1.0, Probability = 0.0515501057576029, Class = 0
Input = 2.0, Probability = 0.16412826954644827, Class = 0
Input = 3.0, Probability = 0.4149888799701166, Class = 0
Input = 4.0, Probability = 0.7193154698480789, Class = 1
Input = 5.0, Probability = 0.9025176083649726, Class = 1

Sumeet S019

Process finished with exit code 0

```